

Li Ju

Uppsala, Sweden | +46 0762733465 | li-ju666@outlook.com | <https://li-ju666.com>

SUMMARY

Final-year Ph.D. candidate working on geometry-aware probabilistic modeling and uncertainty quantification for vision-language foundation models. Research combines generative modeling, representation geometry, and post-hoc adaptation of pre-trained encoders, with publications at NeurIPS and ICML. Seeking research positions in representation learning, generative modeling on manifolds, or probabilistic methods for foundation models.

EDUCATION

Ph.D. in Scientific Computing <i>Uppsala University</i>	09.2021 – 06.2026 (Expected) <i>Uppsala, Sweden</i>
M.Sc. in Computational Science <i>Uppsala University</i>	09.2019 – 08.2021 <i>Uppsala, Sweden</i>
M.Sc. in Chemometrics, with Honors <i>University of Science and Technology of China</i>	09.2016 – 06.2019 <i>Hefei, China</i>
B.Sc. in Chemistry <i>University of Science and Technology of China</i>	09.2012 – 06.2016 <i>Hefei, China</i>

WORK EXPERIENCE

Doctoral researcher <i>Scientific Machine Learning Laboratory</i>	09.2021 – Present <i>Uppsala, Sweden</i>
- Designed and implemented novel algorithms for distributed learning and uncertainty quantification in vision-language models using probabilistic and generative modeling, leading to multiple publications. - Contributed to collaborative research of various topics, including attack/defense mechanisms in distributed ML, hierarchical information management systems, and simulation-based inference. - Instructed graduate courses <i>Data Engineering</i> and <i>Applied Cloud Computing</i> and, supervised four M.Sc. course/thesis projects.	
Machine learning engineer intern <i>Modulai</i>	01.2026 – 04.2026 <i>Stockholm, Sweden</i>
- Engineered a distributed LLM post-training infrastructure on HPC environments using Reinforcement Learning with Verifiable Rewards. - Developed a robust pipeline for data synthesis and LLM fine-tuning using verl and vllm, optimizing for in- and out-of-distribution performance.	
Junior data scientist (Part-time) <i>Scaleout Systems AB</i>	06.2020 – 03.2021 <i>Uppsala, Sweden</i>
- Contributed to FEDn, a production-grade open-source federated learning framework, by architecting and integrating PyTorch support. - Developed proof-of-concept applications in computer vision and fraud detection to secure client adoption and demonstrate framework scalability.	

SKILLS & TOOLS

Modeling: Generative modeling, probabilistic modeling, multi-modal language models, flow matching, diffusion models, contrastive learning, uncertainty quantification, Bayesian inference.

Frameworks: PyTorch, JAX, transformers, verl, vllm, FastAPI, pthread, OpenMP, MPI, CUDA.

Cloud & DevOps: Docker, Apptainer, Kubernetes, OpenStack.

Languages: Python, C/C++, Haskell, Erlang, Lisp/Racket.

SELECTED PROJECT EXPERIENCE

Aleatoric uncertainty quantification for VLMs on the unit hypersphere

- Identified and formalized the asymmetric uncertainty structure between vision and language modalities: high aleatoric ambiguity in text versus low aleatoric, high epistemic structure in images, which prior post-hoc methods had treated symmetrically.
- Proposed AsymVLM, a directional-distribution adapter (von Mises-Fisher and Power Spherical) that models text embeddings probabilistically while preserving deterministic image embeddings, framed as a natural extension of CLIP’s contrastive objective.
- Validated improved cross-modal retrieval accuracy and uncertainty calibration across MS-COCO, Flickr30k, and Conceptual Captions, with downstream applications including robust fine-tuning and “none-of-the-above” zero-shot classification.

Epistemic uncertainty quantification for VLMs via Riemannian flow matching

- Established a theoretical link between embedding density and epistemic uncertainty, motivating negative log-likelihood on the manifold as a principled, scalable proxy for model confidence in pre-trained VLMs.
- Designed REPVLM, a conditional Riemannian flow matching framework that estimates probability density directly on the hyperspherical embedding manifold, avoiding parameter-space sampling entirely.
- Achieved near-perfect Spearman correlation with prediction error in selective classification, and demonstrated downstream utility for out-of-distribution detection and automated curation of large-scale web datasets.

Accelerating fair federated learning

- Reformulated fairness-aware federated learning as a dynamic multi-objective optimization problem, addressing the suboptimal convergence rates of existing first-order formulations.
- Proposed AdaFedAdam, an algorithm with adaptive hyperparameter tuning and normalized client updates, with theoretical guarantees for accelerated convergence and reduced fairness bias.
- Empirically validated Pareto optimality and robustness across standard FL benchmarks under varied data and system heterogeneity.

SELECTED PUBLICATIONS

- **Ju, L.**, Nautiyal, M., Vats, E., & Singh, P. Epistemic Uncertainty Quantification for Pre-trained VLMs via Riemannian Flow Matching. ICML 2026.
- **Ju, L.**, Andersson, M., Fredriksson, S., Glöckner, E., Hellander, A., Vats, E., & Singh, P. Exploiting the Asymmetric Uncertainty Structure of Pre-trained VLMs on the Unit Hypersphere. NeurIPS 2025.
- **Ju, L.**, Zhang, T., Toor, S., & Hellander, A. Accelerating Fair Federated Learning: Adaptive Federated Adam. IEEE Transactions on Machine Learning in Communications and Networking.
- **Ju, L.**, Hellander, A., & Spjuth, O. Federated Learning for Predicting Compound Mechanism of Action Based on Image-data From Cell Painting. Artificial Intelligence in the Life Sciences.
- **Ju, L.**, Singh, P., & Toor, S. Proactive Autoscaling for Edge Computing Systems with Kubernetes. Proceedings of the 14th IEEE/ACM International Conference on Utility and Cloud Computing Companion.

AWARDS & LEADERSHIP

- **1st Place in Europe, 2nd Place Globally** in Huawei Sweden Hackathon (2025): Developed a neural operator for SVD approximation, competing against ~500 teams from across the world.
- **2nd Place** in Huawei Sweden Hackathon (2024): Solved a wireless localization problem using channel charting, competing against over 30 European teams.
- **President** of Society for Industrial and Applied Mathematics (SIAM), Uppsala University Chapter (2024 – 2025). Organized seminars and workshops on numerical analysis and scientific computing.